

Initial Business Insights

Phase 1: Data Understanding & Initial Exploration Report

1. Dataset Overview

Dataset Scale

Metric	Value
Total Records (Flights)	7,079,081
Total Columns	35
Memory Usage	4.68 GB
Duplicate Records	0

Data Types Summary

Data Type	Count
Integer Columns	13
Float Columns	13
String Columns	9
Total Columns	35

Key Observation

The dataset contains over 7 million flight records, making it suitable for enterprise-scale analytics, machine learning, and airport operations intelligence use cases.

2. Data Quality Assessment

Missing Values Analysis

Column	Missing Count	Missing %
cancellation_code	6,982,766	98.64%
actual_elapsed_time	113,814	1.61%
arr_delay	113,814	1.61%
air_time	113,814	1.61%
wheels_on	97,856	1.38%
taxi_in	97,856	1.38%
arr_time	97,854	1.38%
wheels_off	95,734	1.35%

taxi_out	95,734	1.35%
dep_delay	92,970	1.31%
dep_time	92,659	1.31%

Data Quality Findings

- No duplicate records were found.
- Missing values are primarily concentrated in operational flight timing columns.
- Missing operational data is strongly associated with cancelled flights.
- Cancellation code is populated only for cancelled flights, resulting in 98.64% null values.

3. Flight Network Overview

Airline Coverage

Metric	Value
Total Airlines	15

Airlines Included

9E, AA, AS, B6, DL, F9, G4, HA, MQ, NK, OH, OO, UA, WN, YX

Airport Coverage

Metric	Value
Origin Airports	348
Destination Airports	348
Unique Airports	348

Observation

The dataset covers a large airport network across the United States, providing sufficient diversity for route optimization, congestion analysis, and delay prediction.

Route Coverage

Metric	Value
Unique Routes	6,805

Top 10 Busiest Routes

Route	Flights
OGG → HNL	11,642
HNL → OGG	11,634
LAX → SFO	11,526
SFO → LAX	11,499
LGA → ORD	10,971
ORD → LGA	10,969
JFK → LAX	9,915
LAX → JFK	9,915
DCA → BOS	9,658
BOS → DCA	9,657

Observation

Traffic volume is highly concentrated among major metropolitan and inter-island routes.

4. Operational Disruption Analysis

Flight Cancellations

Metric	Value
Cancelled Flights	96,315
Cancellation Rate	1.36%

Flight Diversions

Metric	Value
Diverted Flights	17,499

Observation

Although cancellations and diversions represent a relatively small percentage of flights, they contribute significantly to operational disruptions and passenger dissatisfaction.

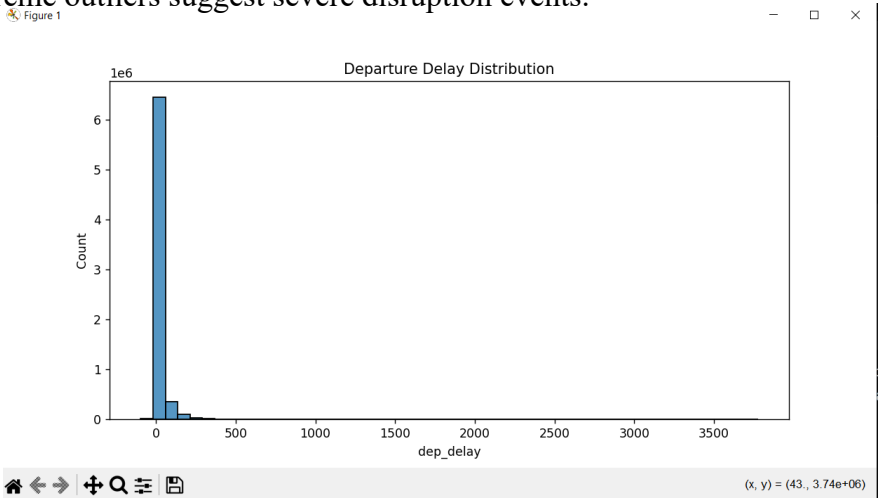
5. Delay Performance Analysis

Departure Delay Statistics

Metric	Value
Average Delay	12.68 min
Median Delay	-2 min
Minimum Delay	-96 min
Maximum Delay	3,777 min

Interpretation

- Negative values indicate early departures.
- Most flights depart close to schedule.
- Extreme outliers suggest severe disruption events.



Arrival Delay Statistics

Metric	Value
Average Delay	7.10 min
Median Delay	-6 min
Minimum Delay	-126 min
Maximum Delay	3,803 min

Interpretation

Arrival performance is generally better than departure performance, indicating some recovery during flight operations.

6. Delay Root Cause Analysis

Total Delay Minutes by Cause

Delay Cause	Total Minutes
Late Aircraft Delay	41,974,007
Carrier Delay	35,823,265
NAS Delay	19,621,994
Weather Delay	6,195,873
Security Delay	179,928

Key Finding

The largest contributor to delays is Late Aircraft Delay, indicating strong delay propagation across the airline network.

Business Implication

A delay affecting one aircraft can trigger cascading delays throughout the day, making delay propagation a critical problem for airport operations.

7. Airport Performance Analysis

Airports with Highest Average Departure Delay

Airport	Average Delay (Minutes)
MGW	72.62
EWN	68.40
HTS	42.79
SMX	40.51
CKB	37.02
OTH	35.35
HGR	35.12
PBG	27.68
HYA	26.64
DEC	26.27

Observation

Certain airports experience significantly higher average delays and may require targeted operational interventions.

8. Airline Performance Analysis

Average Departure Delay by Airline

Airline	Average Delay (Minutes)
AA	20.70
F9	19.00
B6	17.46
NK	14.96
G4	14.05
OH	13.94
UA	11.96
WN	11.75
OO	11.28
DL	10.22
MQ	10.18
9E	8.34
AS	7.63
HA	5.47
YX	3.97

Observation

There is significant variation in operational performance among airlines, providing opportunities for benchmarking and performance monitoring.

9. Congestion Analysis

Peak Scheduled Departure Hours

Highest traffic volume occurs during:

Hour	Flights
06:00	495,017
07:00	493,915
08:00	482,402
17:00	443,706

11:00	442,008
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Observation

Airport operations experience significant congestion during morning and evening departure banks.

Business Implication

Peak-hour congestion likely contributes to taxi delays, departure delays, and delay propagation across the network.

10. Key Business Insights

Insight 1

The dataset represents a large-scale aviation network with over 7 million flights, 348 airports, and 6,805 unique routes.

Insight 2

Late Aircraft Delay is the largest contributor to total delay minutes, indicating that disruption propagation is a major operational challenge.

Insight 3

Departure delays are generally higher than arrival delays, suggesting that airlines recover some lost time during flight operations.

Insight 4

Operational performance varies considerably between airlines, creating opportunities for benchmarking and best-practice analysis.

Insight 5

A small number of airports experience disproportionately high average delays and should be prioritized for operational improvement initiatives.

Insight 6

Morning and evening departure banks represent the highest congestion periods and are likely major drivers of airport operational stress.

Insight 7

Flight cancellations account for approximately 1.36% of all operations, impacting more than 96,000 flights during 2024.